

EMBEDDED SYSTEMS & IoT DEVICE DESIGN

Project Title:

IoT-Based Smart Temperature Monitoring and Alert System

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Table of Content

Sr. No.	Contents	Page No.
1	Abstract	3
2	Introduction to Embedded System	4
3	Applications of Embedded System	5
4	Microcontroller Vs. Microprocessor	6
5	Microcontroller Used in IoT	7
6	Sensor Study	9
7	Project Description	10
8	Arduino Program	12
9	Flowchart	14
10	Results and Observation	15
11	Conclusions and Future Scopes	17

Abstract

The Internet of Things is changing the way we monitor things by letting us collect data in time and access it from anywhere. This project is about making a temperature monitoring system that sends alerts using Arduino and a TMP36 temperature sensor.

The system checks the temperature all the time. Shows the readings on the serial monitor. If the temperature gets too high it sends an alert using a light and a buzzer. This project shows how to use sensors, tiny computers and Internet of Things ideas to make monitoring systems.

We can also use this system with cloud platforms, like ThingSpeak and Blynk to monitor things from away and keep track of the data.

The Internet of Things and temperature monitoring are the focus of this project.

Keywords: Embedded Systems, Internet of Things, Arduino, TMP36 Temperature Monitoring, Smart Sensors.

Introduction to Embedded Systems

❖ What is an Embedded System?

An embedded system is a computer system that is developed for carrying out a particular function in the larger system. An embedded system has hardware parts like microcontrollers, memory, input and output peripherals and software that controls its functions. Embedded systems are known for their reliability, efficiency, size and real-time operations. They are extensively utilized in household appliances including washing machine, microwave oven, digital camera, traffic light control, automobile, and Internet of Things based monitoring system.

❖ Characteristics

1. Dedicated Functionality: They do one thing. Do it well.
2. Real-Time Operation: They work in time giving fast results.
3. Low Power Consumption: They use power.
4. High Reliability: They are very reliable.
5. Compact Design: They are small, in size.
6. Cost Effectiveness: They are made to be affordable.

❖ Architecture of Embedded Systems

Input Devices → Microcontroller → Output Devices

❖ Components:

- Sensors: Help in taking inputs.
- Keypads: Used for giving inputs.
- Displays: Show the output.
- Motors: Do the work.
- LEDs: Give outputs.

❖ Example

In a washing machine, the embedded system controls:

- Water level
- Washing time
- Motor speed
- Rinsing and spinning operations

Embedded System Applications

- **Home Automation**

Smart Lighting

Thermostat Control

Security Systems

Smart Door Locks

Energy Management Systems

- **Automobiles**

Controlled Deployment of Airbags

Anti-lock Braking Systems

Engine Management Systems

Tire Pressure Monitoring System (TPMS)

Automatic Parking Assistance

- **Medical Devices**

ECG Measurement Devices

Blood Pressure Monitoring Systems

Monitoring of Patients

Digital Thermometers

- **Industrial Control Systems**

PLC-Based Systems

Temperature Measurement

Process Automation Systems

Conveyor Belt Control System

Predictive Maintenance Systems

- **Robotics**

Autonomous Robots

Industrial Robots

Drones

Robotic Vacuum Cleaners

Warehouse Automation Robots

- **Consumer Electronics**

Washing Machines

Microwave Ovens

Smart Televisions

Digital Cameras

Air Conditioner

Difference between Microcontroller and Microprocessor

Parameters	Microcontroller	Microprocessor
Definition	Complete computer on a chip	CPU on a chip
Ports	On-chip I/O ports	External I/O ports required
Memory	On-chip RAM and ROM	External RAM and ROM required
Power Consumption	Low	High
Price	Low	High
Speed	Moderate	Very high
Size	Small	Large
Applications areas	Embedded systems, Internet of Things, automation	Computers, laptops, servers
Examples	Arduino UNO, ESP32, PIC	Intel Core i5, AMD Ryzen
Usage purpose	Dedicated to particular tasks	General purpose computing

Microcontrollers Used in IoT

- **Arduino UNO:** The Arduino UNO is an open-source microcontroller development board that operates on the ATmega328P microcontroller. The board has been extensively used in embedded systems and robotics projects for its simplicity, low cost, and easy programming capability.

➤ Key Features

ATmega328P Microcontroller

14 Digital Input/Output Pins

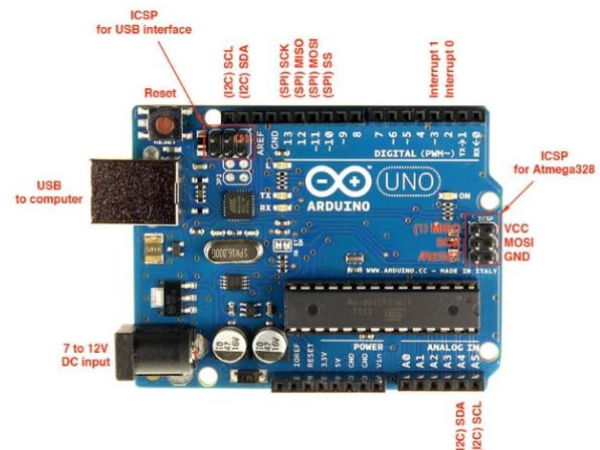
6 Analog Input Pins

32 KB Flash Memory

16 MHz Clock Speed

USB Programming Support

Low Power Consumption



- **ESP32:** ESP32 is a microcontroller that has been made by Espressif Systems. It possesses both WiFi and Bluetooth connectivity capabilities, which makes it suitable for IoT applications.

➤ Key Features

Built-in Wi-Fi and Bluetooth

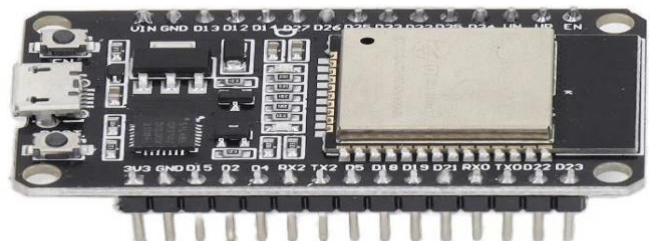
Dual-Core Processor

Low Power Consumption

Multiple GPIO Pins

Analog and Digital Interfaces

High Processing Speed



- **Raspberry Pi Pico:** The Raspberry Pi Pico is an affordable microcontroller board designed by the Raspberry Pi Foundation. The microcontroller used in the Raspberry Pi Pico is known as the RP2040 microcontroller, offering great performance and versatility.

➤ Key Features

RP2040 Dual-Core Microcontroller

264 KB SRAM

26 GPIO Pins

Low Power Consumption

Programmable in C/C++ and MicroPython



Sensors Study

- **Temperature Sensor (TMP36):** The TMP36 is a low-voltage temperature sensor that detects the ambient temperature and generates an output voltage which is proportional to the temperature in degrees Celsius. It is commonly used in embedded systems where the temperature is required to be measured or monitored.

➤ Key Features

Temperature measurement in degree centigrade scale

Low voltage operation (2.7V to 5.5V)

Accuracy up to $\pm 2^{\circ}\text{C}$

Low power consumption

Easy microcontroller interfacing



- **Motion Sensor (PIR):** The Passive Infrared (PIR) Sensor is an electronic sensor that works on detecting movements using infrared rays emanated from human beings and animals.

➤ Key features

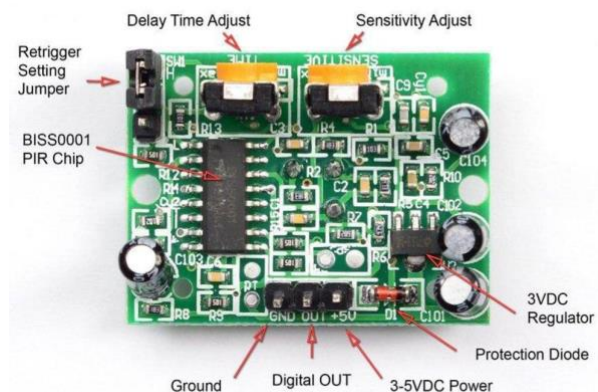
Detects human and animal presence

Uses minimal power

Large detection area

Simple interfacing with microcontrollers

Reliable and economical



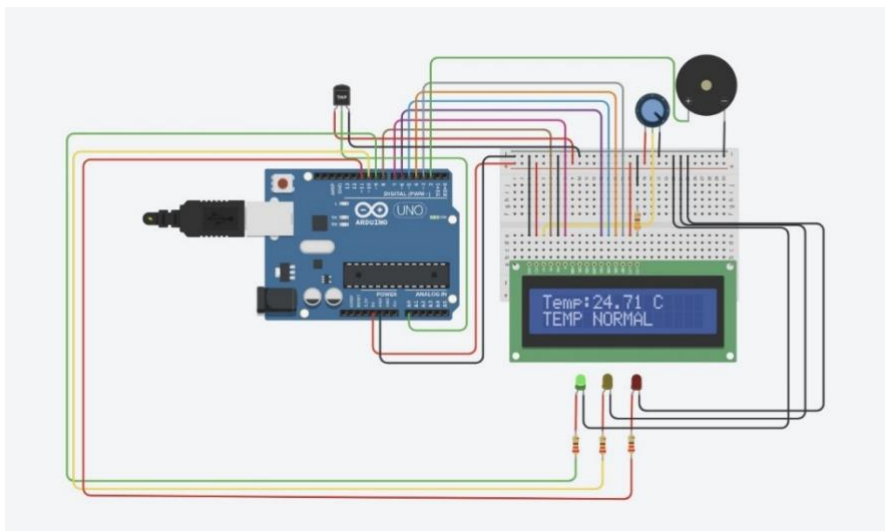
Project Description

- **Aim:** IoT-Based Smart Temperature Monitoring and Alert System
- **Objective:** To constantly keep track of the temperature and to alert in case the temperature goes beyond a certain pre-set value.

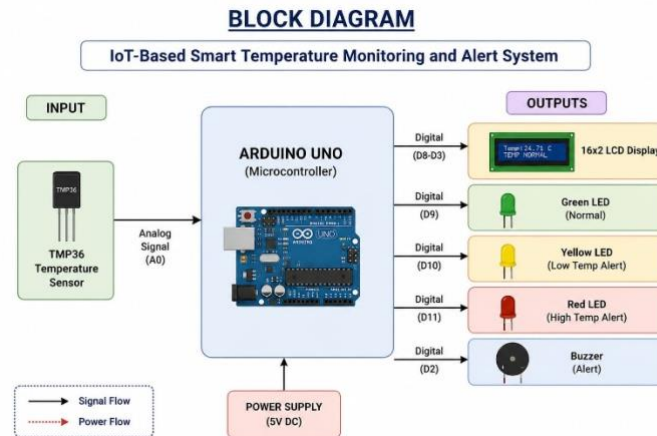
- **Components Used:**

Components	Quantity
Arduino UNO	1
TMP36 Temperature Sensor	1
16×2 LCD Display	1
Piezo Buzzer	1
LED (Red, Yellow, Green)	1 each
220 Ω Resistors	3
Breadboard	1
USB Cable	1
Jumper wires	As required

- **Software Used :** Tinkercad Circuits – For circuit design and simulation.
- **Circuit Diagram :**



- **Block Diagram:**



- **Working Principle :**

1. The TMP36 temperature sensor continuously monitors the temperature of the surrounding environment.
2. The sensor generates a signal from the sensed temperature and passes it to the Arduino UNO through analog pin A0.
3. Arduino UNO receives the analog signal and processes it into the temperature value in degree Celsius (°C).
4. The current temperature value appears on the 16x2 LCD.
5. The Arduino compares the sensed temperature value with the preset temperature limits.
6. When the sensed temperature value lies in the normal limit, the Green LED gets switched ON to show safe operating conditions.
7. When the sensed temperature reaches warning limits, the Yellow LED gets switched ON to indicate the rising temperature.
8. When the sensed temperature goes beyond the critical limit, the Red LED gets switched ON along with the buzzer which generates a sound alarm.
9. As the temperature becomes normal again, the buzzer gets turned OFF and the LED is turned ON as per temperature level and the temperature value appears on the LCD.
10. The Arduino continues doing the above-mentioned process continuously.

Arduino Program :

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(8,7,6,5,4,3);

int sensorPin = A0;
int buzzerPin = 2;

int greenLED = 9;
int yellowLED = 10;
int redLED = 11;

float lowThreshold = 20.0;
float highThreshold = 35.0;

void setup()
{
  Serial.begin(9600);
  lcd.begin(16,2);

  pinMode(buzzerPin, OUTPUT);

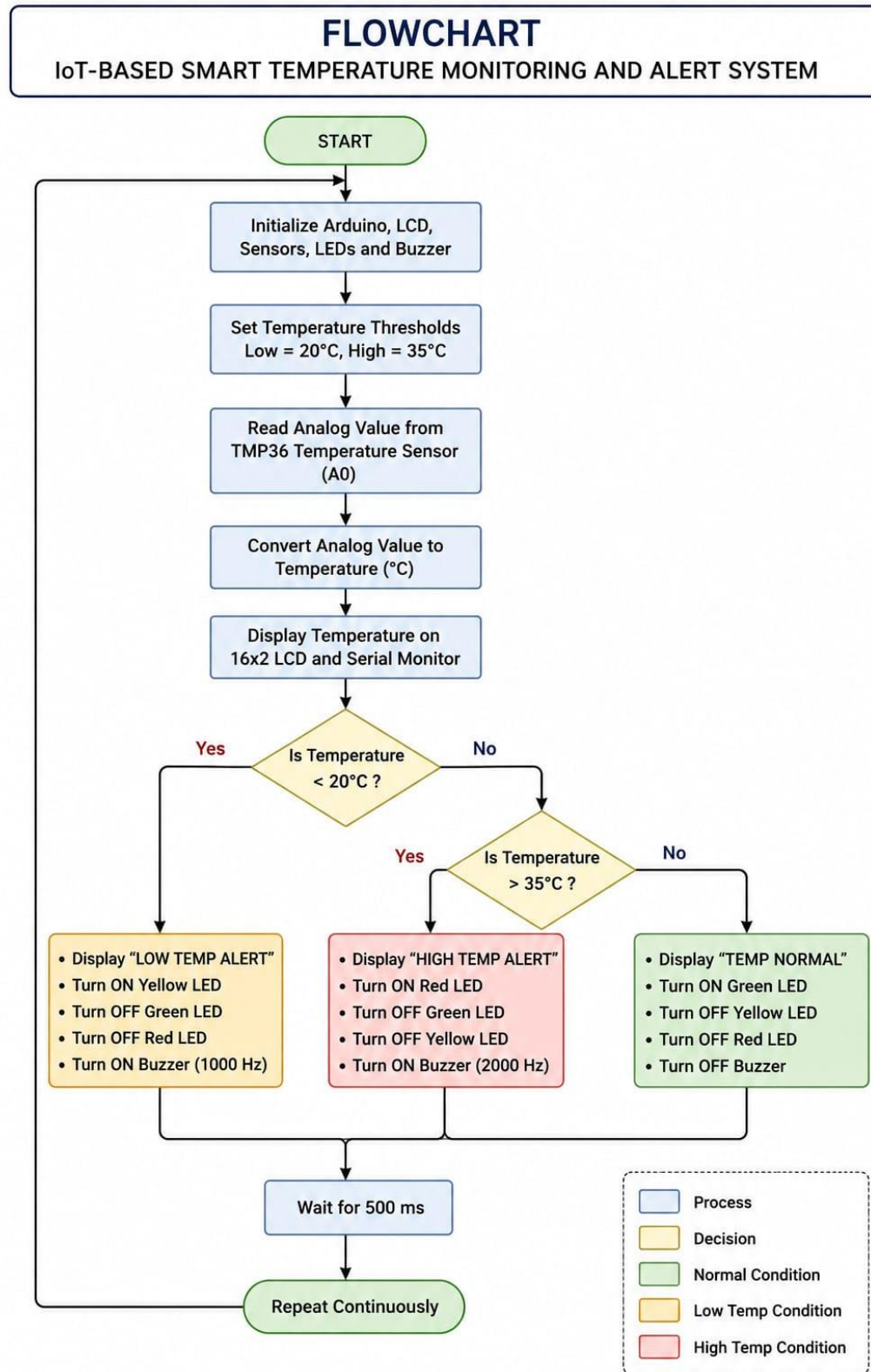
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(redLED, OUTPUT);
}

void loop()
{
  int reading = analogRead(sensorPin);
  float voltage = reading * 5.0 / 1024.0;
  float temperatureC = (voltage - 0.5) * 100.0;
  Serial.print("Temperature: ");
  Serial.println(temperatureC);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Temp:");
  lcd.print(temperatureC);
  lcd.print(" C");
  if (temperatureC < lowThreshold)
  {
    lcd.setCursor(0,1);
    lcd.print("LOW TEMP ALERT");
    digitalWrite(yellowLED, HIGH);
    digitalWrite(greenLED, LOW);
    digitalWrite(redLED, LOW);
    tone(buzzerPin, 1000);
  }
  else if (temperatureC > highThreshold)
  {
    lcd.setCursor(0,1);
    lcd.print("HIGH TEMP ALERT");
    digitalWrite(redLED, HIGH);
    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, LOW);
    tone(buzzerPin, 2000);
  }
  else
  {
    lcd.setCursor(0,1);
    lcd.print("TEMP NORMAL");
    digitalWrite(greenLED, HIGH);
    digitalWrite(yellowLED, LOW);
    digitalWrite(redLED, LOW);
    noTone(buzzerPin);
  }
  delay(500);
}
```

➤ **Algorithm :**

1. Start the program.
2. Initialize the Serial Monitor and 16×2 LCD.
3. Configure the TMP36 sensor, buzzer, and Green, Yellow, and Red LEDs as input/output devices.
4. Read the analog value from the TMP36 temperature sensor through pin A0.
5. Convert the analog value into temperature in degrees Celsius (°C).
6. Display the measured temperature on the LCD and Serial Monitor.
7. Compare the measured temperature with the predefined threshold values:
8. If temperature < 20°C:
 Display "LOW TEMP ALERT" on the LCD.
 Turn ON Yellow LED.
 Turn OFF Green and Red LEDs.
 Activate the buzzer at 1000 Hz.
9. Else if temperature > 35°C:
 - Display "HIGH TEMP ALERT" on the LCD.
 - Turn ON Red LED.
 - Turn OFF Green and Yellow LEDs.
 - Activate the buzzer at 2000 Hz.
10. Otherwise (20°C to 35°C):
 - Display "TEMP NORMAL" on the LCD.
 - Turn ON Green LED.
 - Turn OFF Yellow and Red LEDs.
 - Turn OFF the buzzer.
11. Wait for 500 milliseconds.
12. Repeat the process continuously for real-time temperature monitoring.
13. End (the loop continues until the system is powered off).

Flowchart

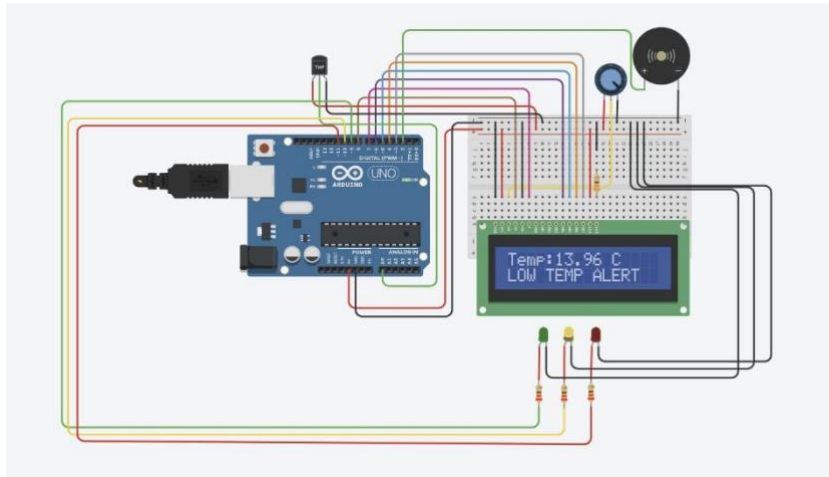


Results and Observations

Output 1:

Temperature Below Threshold(<20)

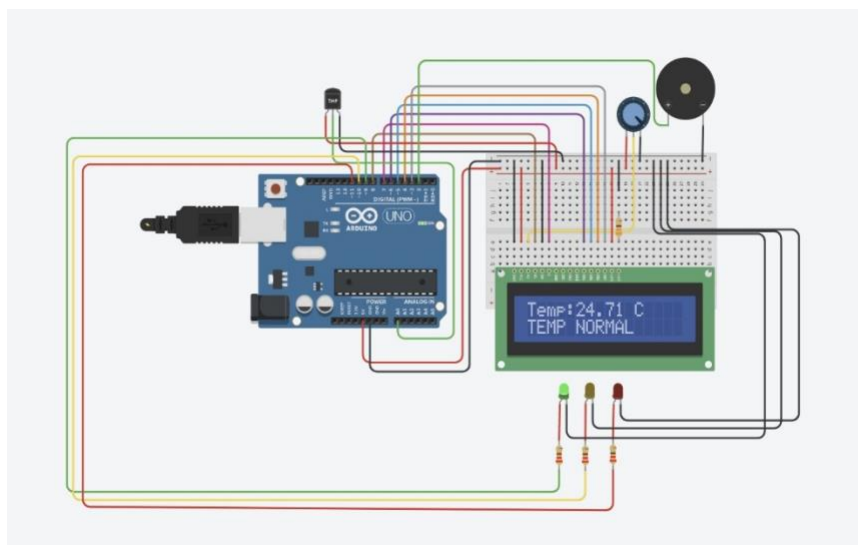
- Yellow color LED ON
- Buzzer ON
- Low temperature alert



Output 2 :

Temperature between Threshold(20°C- 35°C)

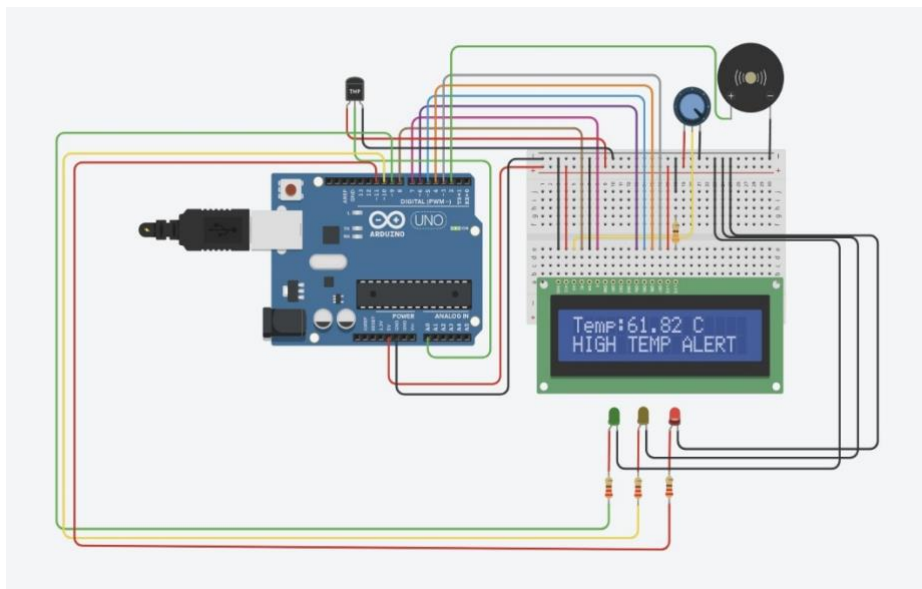
- Green color LED ON
- Buzzer OFF
- Normal temperature



Output 3 :

Temperature Above Threshold(>35)

- RED color LED ON
- Buzzer ON
- High temperature alert



• Observations

The system was able to detect changes in temperature and issue an alarm once the threshold level was breached.

• Advantages:

1. Low-cost implementation
2. Easy to build
3. Real time monitoring
4. Reliable temperature alerts
5. Expandable for IOT applications

• Applications:

1. Hospitals
2. Cold storage
3. Greenhouse
4. Industries
5. Data centers
6. Smart homes
7. laboratories

Conclusion and Future Scopes

- **Conclusion:**

IoT-Based Smart Temperature Monitoring and Alerting System was implemented and tested successfully. It monitors the temperature of the environment and sends an alert whenever the temperature goes beyond the specified threshold.

It is a great example of how embedded systems, sensors, and Internet of Things technology can be used practically.

- **Future Scopes :**

1. Cloud Integration using ThingSpeak
2. App Monitoring on Mobile
3. SMS Alerts
4. LCD Display
5. Smart Fan System
6. Industrial Automation
7. Applications in Smart Agriculture

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